Linear Algebra (ch. 0h)

Linear Algebra starts with systems of linear equations, e.g:

Find x and y satisfying

$$x - 2y = 1 \quad (1)$$
$$2x - 7y = -7 \quad (2)$$

Eliminate variables

$2 \times (1)$: $2x - 4y = 2$

$(2) - 2\times(1)$: $-3y = -9$

$y = 3$

$x = 1 + 2y = 7$

Example 2: Find x, y, z satisfying

$$x + 2y + 2z = 1 \quad (1)$$
$$x + y + z = 1 \quad (2)$$

$(1) - (2)$: $y + z = 0$

$z = -y$

Sub into (2)

$x = 1 - y - z$

$= 1 - y + y$

$x = 1$

Infinitely many solutions : $(x, y, z) = (1, \lambda, -\lambda) \quad \lambda \in \mathbb{R}$

$\rightarrow$ parametric form of a line in $\mathbb{R}^3$

A system of linear equations with 3 variables can geometrically only be one of the following:

- empty set - no solution
- A point in $\mathbb{R}^3$ - unique solution
- A line in $\mathbb{R}^3$
- A plane in $\mathbb{R}^3$
- All of $\mathbb{R}^3$.

What is a matrix?

A rectangular array of numbers :)

More precisely, an $m \times n$ matrix is a rectangular array of entries arranged in m rows and n columns,

e.g $\begin{pmatrix} 6 & 5 & 4 \\ 3 & 2 & 1 \end{pmatrix}$ is a 2×3 matrix.

We use $Mat_{m,n}(\mathbb{R})$ to be the set of all real valued matrices of size $m \times n$

Note $Mat_n(\mathbb{R})$ is the set of square real valued matrix of size $n \times n$

A vector is a matrix with just one column.

$$\left.\begin{array}{l} \text{e.g } \underline{u} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \text{ is a 2-d vector in } \mathbb{R}^2 \\ \underline{v} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{pmatrix} \text{ is a 5-d vector in } \mathbb{R}^5 \end{array}\right\}$$

These might be called column vectors to distinguish them from row vectors (matrices with just one row).

Using example equations from before and turning them into matrices

$$\begin{matrix} x - 2y = 1 \\ 2x - 7y = -7 \end{matrix} \longrightarrow \underbrace{\begin{pmatrix} 1 & -2 \\ 2 & -7 \end{pmatrix}}_{\text{Coefficients}} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ -7 \end{pmatrix}$$

Matrix Operations

Vectors can be added by adding them componentwise

e.g. $\begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 3 \\ -5 \end{pmatrix} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$

Matrix addition : let $A = (a_{ij})$ and $B = (b_{ij})$ with $A, B \in Mat_{n,m}(\mathbb{R})$

Then $\underset{(c_{ij})}{\overset{}{C}} = A + B \in Mat_{n,m}(\mathbb{R})$ with $c_{ij} = a_{ij} + b_{ij}$

Scalar multiplication : $\underline{a} = \begin{pmatrix} 2 \\ 5 \\ 7 \end{pmatrix}$, $5\underline{a} = \begin{pmatrix} 10 \\ 25 \\ 35 \end{pmatrix}$

Similarly for an $n \times m$ matrix $A = (a_{ij}) \in Mat_{n,m}(\mathbb{R})$ and $\lambda \in \mathbb{R}$, $\lambda \cdot A$ is the matrix of size $n \times m$ with entries $\lambda \cdot A = (\lambda \cdot a_{ij})$

Matrix-vector multiplication:

$$\begin{matrix} x + 2y + 2z = 1 \\ x + y + z = 1 \end{matrix} \qquad A = \underbrace{\begin{pmatrix} 1 & 2 & 2 \\ 1 & 1 & 1 \end{pmatrix}}_{\text{Coefficients}} \qquad \underline{b} = \underbrace{\begin{pmatrix} 1 \\ 1 \end{pmatrix}}_{\text{Constants}} \qquad \underline{v} = \underbrace{\begin{pmatrix} x \\ y \\ z \end{pmatrix}}_{\text{Variables}}$$

$$A \cdot \underline{v} = \underline{b}$$

$$\begin{pmatrix} 1 & 2 & 2 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

To get back to the linear system, we need to multiply rows of A with the column of $\underline{v}$.

$$\begin{pmatrix} \boxed{a \quad b \quad c} \\ \boxed{d \quad e \quad f} \end{pmatrix} \begin{pmatrix} \boxed{\begin{matrix} x \\ y \\ z \end{matrix}} \end{pmatrix} = \begin{pmatrix} g \\ h \end{pmatrix}$$

$ax + by + cz = g$

$dx + ey + fz = h$

In general:

$$\underbrace{\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{pmatrix}}_{\in Mat_{n \times m}(\mathbb{R})} \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix}}_{\in Mat_{m \times 1}(\mathbb{R}) = \mathbb{R}^m} = \underbrace{\begin{pmatrix} a_{11}x_1 + a_{12}x_2 + \dots + a_{1m}x_m \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2m}x_m \\ \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nm}x_m \end{pmatrix}}_{\in Mat_{n \times 1}(\mathbb{R}) = \mathbb{R}^n}$$

$$A: \mathbb{R}^m \to \mathbb{R}^n$$
$$\underline{x} \mapsto A\underline{x}$$

→ multiplying a vector $\underline{v} \in \mathbb{R}^m$ by a matrix $A \in Mat_{n\times m}(\mathbb{R})$ produces a vector $\underline{w} \in \mathbb{R}^n$

It maps an m-d vector to an n-d vector.

Matrix - Matrix multiplication:

$$A \in Mat_{n\times \underline{m}}(\mathbb{R}), \quad B \in Mat_{\underline{m}\times r}(\mathbb{R})$$

→ $A \cdot B \in Mat_{m\times r}(\mathbb{R})$ by multiplying rows of A by columns of B.

Example:

$$A \cdot B = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} 5 & 6 & 7 \\ 8 & 9 & 10 \end{pmatrix} = \begin{pmatrix} 1\times5+2\times8 & 1\times6+2\times9 & 1\times7+2\times10 \\ 3\times5+4\times8 & 3\times6+4\times9 & 3\times7+4\times10 \end{pmatrix}$$

$$= \begin{pmatrix} 21 & 24 & 27 \\ 47 & 54 & 61 \end{pmatrix}$$

to multiply A with B, the row length of A must be equal to the column length of B.

$$A \in Mat_{n\times \underline{m}}(\mathbb{R}), \quad B \in Mat_{\underline{m}\times r}(\mathbb{R})$$

must be equal

Resulting matrix $AB \in Mat_{n\times r}(\mathbb{R})$

- Even if BA and AB exist, they are not always equal

e.g.
$$\begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 6 & 9 \end{pmatrix}$$
$$\neq$$
$$\begin{pmatrix} 0 & 1 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 7 & 8 \end{pmatrix}$$

- Just because $A^2 = \begin{pmatrix} 0 & 0 & \cdots & 0 \\ 0 & & \ddots & \vdots \\ \vdots & & & \\ 0 & \cdots & & 0 \end{pmatrix}$ does not mean $A = \begin{pmatrix} 0 & 0 & \cdots & 0 \\ 0 & & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \cdots & & 0 \end{pmatrix}$

E.g. $A = \begin{pmatrix} 2 & 4 \\ -1 & -2 \end{pmatrix} \neq 0$

but $A^2 = \begin{pmatrix} 2 & 4 \\ -1 & -2 \end{pmatrix}\begin{pmatrix} 2 & 4 \\ -1 & -2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = 0$

We can think of a matrix $A \in Mat_{n\times m}(\mathbb{R})$ as a map from $\mathbb{R}^m$ to $\mathbb{R}^n$ via

$$\underline{x} \in \mathbb{R}^m \xrightarrow[A]{} A\underline{x} \in \mathbb{R}^n$$

The matrix $B \in Mat_{m\times r}(\mathbb{R})$ is a map from $\mathbb{R}^r$ to $\mathbb{R}^m$ via

$$\underline{y} \in \mathbb{R}^r \xrightarrow[B]{} B\underline{y} \in \mathbb{R}^m$$

The matrix $A \cdot B$ is then a map from $\mathbb{R}^r$ to $\mathbb{R}^n$ which is nothing but the composition of the map of B followed by the map of A:

$$\begin{tikzcd} \mathbb{R}^r \arrow[r, "B"] \arrow[rr, bend right, "A\cdot B"'] & \mathbb{R}^m \arrow[r, "A"] & \mathbb{R}^n \end{tikzcd}$$

- Identity Matrix:

$$I = I_n = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & & 0 \\ 0 & 0 & 1 & & \vdots \\ \vdots & \vdots & & \ddots & \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix} \in Mat_n(\mathbb{R})$$

Then $A \cdot I = A = I \cdot A$

. . rix Transposition:

Given $A = (a_{ij}) \in Mat_{n\times m}(\mathbb{R})$, the transpose $A^T \in Mat_{m\times n}(\mathbb{R})$ is obtained by swapping rows and columns, $A^T = (a_{ji})$

E.g. $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}^T = \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}$$

Matrices $A \in Mat_n(\mathbb{R})$ satisfying $A^T = A$ are called symmetric

Important: $(A\cdot B)^T = B^T \cdot A^T$

$$(A\cdot B\cdot C\cdot D)^T = D^T\cdot C^T\cdot B^T\cdot A^T$$

Back to $A\underline{x} = \underline{b}$:

If there were a matrix A^{-1}, we could left multiply by A^{-1} to get x

$$A^{-1}(A\underline{x}) = A^{-1}\underline{b}$$

$$(A^{-1}A)\underline{x} = A^{-1}\underline{b}$$

$$\underline{I}\,\underline{x} = A^{-1}\underline{b} = x$$

$$\Rightarrow \underline{x} = A^{-1}\underline{b}$$

Elementary Row Operations and Gaussian Elimination

Given a system of linear equations, we are allowed to carry out the following operations without changing the solutions:

(S) Swap two equations

(M) Multiply an equation by a non-zero constant

(A) Add a multiple of one equation to another equation

Gaussian Elimination, or row reduction, is an efficient and systematic way to simplify a set of equations, eliminating variables to solve the system

We focus on 3×3 systems:

$$\left.\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 \end{aligned}\right\} \quad A\underline{x} = \underline{b}$$

Introduce the augmented matrix $(A|\underline{b})$:

$$\left(\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{array}\right)$$

- Each row corresponds to an equation.
- SMA operations correspond to operations of the rows of $(A|\underline{b})$
- write R_i for the i^{th} row of $(A|\underline{b})$
  - $R_i \leftrightarrow R_j$ = swap rows i and j
  - cR_i = multiply row i by constant c, $c \neq 0$
  - $R_i + cR_j$ = add c times R_j to R_i

Apply these operations until we have a triangular shape, i.e.:

$$(A|\underline{b}) \to \left(\begin{array}{ccc|c} 1 & a & b & d \\ 0 & 1 & c & e \\ 0 & 0 & 1 & f \end{array}\right)$$

$\hookrightarrow x_3 = f$

$x_2 + cx_3 = e \to x_2 + cf = e \to x_2 = e - cf$

$x_1 + ax_2 + bx_3 = d \to x_1 = d - a(e-cf) - bf$

Example: Solve
$$\begin{cases} 2x + 5y - 4z = -13 \\ x + y + 2z = 7 \\ x + 2y - z = -3 \end{cases}$$

$$\hookrightarrow \left(\begin{array}{ccc|c} 2 & 5 & -4 & -13 \\ 1 & 1 & 2 & 7 \\ 1 & 2 & -1 & -3 \end{array}\right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 7 \\ 2 & 5 & -4 & -13 \\ 1 & 2 & -1 & -3 \end{array}\right)$$

$$\xrightarrow{R_2 - 2R_1,\ R_3 - R_1} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 7 \\ 0 & 3 & -8 & -27 \\ 0 & 1 & -3 & -10 \end{array}\right) \xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 7 \\ 0 & 1 & -3 & -10 \\ 0 & 3 & -8 & -27 \end{array}\right)$$

$$\xrightarrow{R_3 - 3R_2} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 7 \\ 0 & 1 & -3 & -10 \\ 0 & 0 & 1 & 3 \end{array}\right) \to$$

$x + y + 2z = 7$

$y - 3z = -10$

$\underline{z = 3}$

$y = -10 + 3 \times 3 = -1$

$x = 7 + 1 - 6 = 2$

$\therefore x = 2,\ y = -1,\ z = 3$

Example: Solve $\begin{cases} x+2y+3z=1 \\ 4x+5y+6z=1 \\ 7x+8y+9z=5 \end{cases}$

$$\hookrightarrow \left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 4 & 5 & 6 & 1 \\ 7 & 8 & 9 & 5 \end{array}\right) \xrightarrow[R_3-7R_1]{R_2-4R_1} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & -3 & -6 & -3 \\ 0 & -6 & -12 & -12 \end{array}\right)$$

$$\downarrow -\tfrac{1}{6}R_3 \quad -\tfrac{1}{3}R_2$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & 2 & 1 \\ \boxed{0 & 0 & 0 & 1} \end{array}\right) \xleftarrow{R_3-R_2} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 1 & 2 & 2 \end{array}\right)$$

$0x+0y+0z=1$

uh oh

System has no solutions.

Example: Solve $\begin{cases} x+2y+3z=1 \\ 4x+5y+6z=1 \\ 7x+8y+9z=1 \end{cases}$

$$\hookrightarrow \left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 4 & 5 & 6 & 1 \\ 7 & 8 & 9 & 1 \end{array}\right) \xrightarrow[R_3-7R_1]{R_2-4R_1} \left(\begin{array}{cccc} 1 & 2 & 3 & 1 \\ 0 & -3 & -6 & -3 \\ 0 & -6 & -12 & -6 \end{array}\right)$$

$$\downarrow -\tfrac{1}{6}R_3 \quad -\tfrac{1}{3}R_2$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & 2 & 1 \\ \boxed{0 & 0 & 0 & 0} \end{array}\right) \xleftarrow{R_3-R_2} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 1 & 2 & 1 \end{array}\right)$$

$0x+0y+0z=0$

$$\hookrightarrow \begin{aligned} x+2y+3z&=1 \\ y+2z&=1 \end{aligned}$$

Choose for $z=\lambda\in\mathbb{R}$

$$\begin{aligned} \to y &= 1-2\lambda \\ x &= 1-2(1-2\lambda)-3\lambda \\ &= -1+\lambda \end{aligned}$$

$\therefore$ Solution set $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -1+\lambda \\ 1-2\lambda \\ \lambda \end{pmatrix} = \underbrace{\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}}_{\text{This is geometrically a line in } \mathbb{R}^3}$

RREF → reduced row echelon form

échelon  → ladder/staircase

Example (non-square):

$$x + 2y + 3z = 1$$
$$4x + 5y + 6z = 1$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 4 & 5 & 6 & 1 \end{array}\right) \xrightarrow{R_2 - 4R_1} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & -3 & -6 & -3 \end{array}\right) \xrightarrow{-\frac{1}{3}R_2} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & 2 & 1 \end{array}\right)$$

$$x + 2y + 3z = 1$$
$$y + 2z = 1$$

we call z a free variable - can have any value $\lambda \in \mathbb{R}$.

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{matrix} -1+\lambda \\ 1-2\lambda \\ \lambda \end{matrix} \qquad \lambda \in \mathbb{R}$$

Example 2:

$$x + 2y + 3z = 3$$
$$2x + y - z = -3$$
$$x + y + z = 4$$
$$2x + 3y + 3z = 7$$

More equations than variables

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 2 & 1 & -1 & -3 \\ 1 & 1 & 2 & 4 \\ 2 & 3 & 3 & 7 \end{array}\right) \xrightarrow[\substack{R_2 - 2R_1 \\ R_3 - R_1 \\ R_4 - 2R_1}]{} \left(\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 0 & -3 & -3 & -9 \\ 0 & -1 & 1 & 1 \\ 0 & -1 & 1 & 1 \end{array}\right) \xrightarrow[R_4 - R_3]{-\frac{1}{3}R_2} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 3 \\ 0 & 1 & 1 & 3 \\ 0 & -1 & -1 & 1 \\ 0 & 0 & -1 & 0 \end{array}\right)$$

$$\xrightarrow{R_3 + R_2} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 3 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{array}\right) \xrightarrow{\frac{1}{2}R_3} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 3 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

↳ $x + 2y + 3z = 3$

$y + z = 3$

$z = 2$

$(x, y, z) = (-1, 1, 2)$

RREF example:

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & -3 \\ 0 & 1 & -3 & -10 \\ 0 & 0 & 1 & 3 \end{array}\right) \xrightarrow[R_2+3R_3]{R_1+R_3} \left(\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right) \xrightarrow{R_1-2R_2} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right)$$

$$\underbrace{\begin{matrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{matrix}}_{I_3}$$

↳ $x=2$, $y=-1$, $z=3$

A matrix A of any size is said to be in Row reduced echelon form (rref) if:

(i) The leftmost non-zero entry in a non-zero row is a 1 → a leading 1

(ii) If a row has a leading 1, then:

(a). all other entries in its column are 0

(b). Leading 1s in lower rows are to the right (staircase pattern)

(iii) Rows of all 0s must be below all other rows.

Examples:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 1 & 4 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 4 \end{pmatrix} \quad \left(\begin{array}{cccc|c} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & 3 & 0 & 2 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 4 \end{array}\right)$$

↳ this would have 0 Solutions
— $0x+0y+0z+0w=4$ ✗

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & 2 & 3 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

↳ 2 free variables
$z=\lambda$, $w=\mu$

$y = 2-2z-3w \qquad \lambda,\mu \in \mathbb{R}$
$= 2-2\lambda-3\mu$

$x=-1$

$$\begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} -1 \\ 2-2\lambda-3\mu \\ \lambda \\ \mu \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ -2 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ -3 \\ 0 \\ 1 \end{pmatrix}$$

↳ this solution set represents a plane.

Theorem: We can always use EROs to turn any matrix A into RREF.

Structure of Solutions of linear Systems

There are different sets of solutions to $(A|b)$ $A\underline{x}=b$

① Having a unique solution:

$$\left(\begin{array}{cccc|c} 1 & 0 & \cdots & 0 & x \\ 0 & 1 & \cdots & 0 & \vdots \\ \vdots & & \ddots & \vdots & \vdots \\ 0 & \cdots & 1 & 0 & x \\ 0 & \cdots & \cdots & 0 & 0 \\ 0 & \cdots & \cdots & 0 & 0 \\ \vdots & & & \vdots & \vdots \\ 0 & \cdots & \cdots & 0 & 0 \end{array}\right)$$

all 0-rows must equal 0 on RHS

RREF has identity matrix followed by zero rows

② No Solution :

$$\left(\begin{array}{ccc|c} \vdots & & \vdots & \vdots \\ 0 & 0 \cdots & 0 & x \neq 0 \\ \vdots & & \vdots & \vdots \end{array}\right)$$

RREF must have a row with all 0s on LHS but a non-zero on the RHS

③ Infinitely many solutions

$$\left(\begin{array}{cccc|c} 1 & 0 & \cdots & 0 & \\ & 1 & \text{Free vars} & \vdots & \\ & & 1 \; \boxed{x \;\; x} & \vdots & \\ & & 1 & 0 & \\ & & \ddots & 1 & \\ 0 & \cdots & \cdots & 0 & 0 \\ \vdots & & & \vdots & \vdots \end{array}\right)$$

This happens when number of rows with leading 1s in RREF is less than the number of columns in the matrix and no inconsistency in the bottom rows → more variables than equations + there is a solution.

For $A\underline{x} = \underline{b}$, then $A\underline{x} = 0$ is the associated homogeneous system of linear equations. $A\underline{x} = 0$ has always $\underline{x} = \underline{0}$ as a solution so any homogeneous equation $A\underline{x} = 0$ has at least the solution $\underline{x} = 0$, but may have infinitely many solutions

Example: $x + y = 0$
$2x + 2y = 0$

$$\left(\begin{array}{cc|c} 1 & 1 & 0 \\ 2 & 2 & 0 \end{array}\right) \xrightarrow{R_2 - 2R_1} \left(\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right)$$

y is a free variable
⟹ Infinitely many solutions

$$\begin{pmatrix} x \\ y \end{pmatrix} = \underline{\begin{pmatrix} -\lambda \\ \lambda \end{pmatrix}} \quad \lambda \in \mathbb{R}$$

A related inhomogeneous equation:
$x + y = 1$
$2x + 2y = 2$

$$\left(\begin{array}{cc|c} 1 & 1 & 1 \\ 2 & 2 & 2 \end{array}\right) \xrightarrow{R_2 - 2R_1} \left(\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 0 & 0 \end{array}\right) \qquad x = 1 - y$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1-\lambda \\ \lambda \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

Generally, we have:

Given any inhomogeneous equation $A\underline{x} = \underline{b}$ with $\underline{b} \neq \underline{0}$, Suppose we have two solutions $\underline{x}_1$, $\underline{x}_2$, i.e. $A\underline{x}_1 = \underline{b}$, $A\underline{x}_2 = \underline{b}$, then $A(\underline{x}_1 - \underline{x}_2) = A\underline{x}_1 - A\underline{x}_2 = \underline{b} - \underline{b} = \underline{0}$, this means that the difference of any two solutions of the original inhomogeneous equation is a solution of the homogeneous equation $A\underline{x} = \underline{0}$.

I.e. If we choose one particular solution $\underline{x}_p$ to $A\underline{x}_p = \underline{b}$, then all solutions of $A\underline{x} = \underline{b}$ are of the form $\underline{x} = \underline{x}_p + \underline{x}_h$, where $\underline{x}_h$ is an arbitrary solution of $A\underline{x} = \underline{0}$

Linear Systems With a parameter

Example:
$$\begin{cases} x + y + z = \alpha + 1 \\ x + \alpha y + z = 1 \\ \alpha x + y + (\alpha - 1) z = \alpha \end{cases}$$

for which $\alpha \in \mathbb{R}$ do we have a unique solution, no solution or infinitely many solutions

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & \alpha+1 \\ 1 & \alpha & 1 & 1 \\ \alpha & 1 & \alpha-1 & \alpha \end{array}\right) \xrightarrow[R_3-\alpha R_1]{R_2-R_1} \left(\begin{array}{ccc|c} 1 & 1 & 1 & \alpha+1 \\ 0 & \alpha-1 & 0 & -\alpha \\ 0 & 1-\alpha & -1 & \alpha-\alpha(\alpha+1)=-\alpha^2 \end{array}\right)$$

$$\downarrow R_3+R_2$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & \alpha+1 \\ 0 & \alpha-1 & 0 & -\alpha \\ 0 & 0 & -1 & -\alpha^2-\alpha \end{array}\right)$$

Assuming $\alpha=1$, the second row leads to an inconsistency: $0=-1$, and we have no solution

$\alpha \neq 1$

$$\xrightarrow[-R_3]{\frac{1}{\alpha-1}R_2} \left(\begin{array}{ccc|c} 1 & 1 & 1 & \alpha+1 \\ 0 & 1 & 0 & -\frac{\alpha}{\alpha-1} \\ 0 & 0 & 1 & \alpha^2+\alpha \end{array}\right) \xrightarrow{R_1-R_2} \left(\begin{array}{ccc|c} 1 & 0 & 1 & \alpha+1+\frac{\alpha}{\alpha-1} \\ 0 & 1 & 0 & -\frac{\alpha}{\alpha-1} \\ 0 & 0 & 1 & \alpha^2+\alpha \end{array}\right)$$

$$\downarrow R_1-R_3$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & \alpha+1+\frac{\alpha}{\alpha-1}-\alpha-\alpha^2 \\ 0 & 1 & 0 & -\frac{\alpha}{\alpha-1} \\ 0 & 0 & 1 & \alpha^2+\alpha \end{array}\right)$$

$$\downarrow$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1-\alpha^2+\frac{\alpha}{\alpha-1} \\ 0 & 1 & 0 & -\frac{\alpha}{\alpha-1} \\ 0 & 0 & 1 & \alpha^2+\alpha \end{array}\right)$$

$\therefore$ Unique solution is $\begin{pmatrix} 1-\alpha^2+\frac{\alpha}{\alpha-1} \\ \frac{-\alpha}{\alpha-1} \\ \alpha^2+\alpha \end{pmatrix}$

for $\alpha \neq 1$

Example

$$x + y + z = -1$$
$$tx + y + z = -1$$
$$x + t^2 y + z = t$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & -1 \\ t & 1 & 1 & -1 \\ 1 & t^2 & 1 & t \end{array}\right) \xrightarrow[R_3 - R_1]{R_2 - tR_1} \left(\begin{array}{ccc|c} 1 & 1 & 1 & -1 \\ 0 & 1-t & 1-t & t-1 \\ 0 & t^2-1 & 0 & t+1 \end{array}\right)$$

Assuming $t = 1$: $\left(\begin{array}{ccc|c} 1 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 \end{array}\right)$ No Solutions for $t = 1$

For $t \neq 1$ $(1-t \neq 0)$:

$$\xrightarrow{\frac{1}{1-t}R_2} \left(\begin{array}{ccc|c} 1 & 1 & 1 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & t^2-1 & 0 & t+1 \end{array}\right) \xrightarrow{R_1 - R_2} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & -1 \\ 0 & t^2-1 & 0 & t+1 \end{array}\right)$$

$t^2 - 1 = (t+1)(t-1)$

If $t = -1$, we have:

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right) \Rightarrow \text{Infinitely many solutions } \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ -1-\lambda \\ \lambda \end{pmatrix}, \ \lambda \in \mathbb{R}, \text{ for } t = -1$$

If $t \neq -1$, then $t^2 - 1 \neq 0$

$$\xrightarrow{\frac{1}{t^2-1}R_3} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & -1 \\ 0 & 1 & 0 & \frac{t+1}{t^2-1} = \frac{1}{t-1} \end{array}\right) \xrightarrow{R_2 - R_3} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 - \frac{1}{t-1} \\ 0 & 1 & 0 & \frac{1}{t-1} \end{array}\right) \xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{t-1} \\ 0 & 0 & 1 & -1 - \frac{1}{t-1} \end{array}\right)$$

$$\Rightarrow \text{Unique solution } \begin{pmatrix} 0 \\ \frac{1}{t-1} \\ -1 - \frac{1}{t-1} \end{pmatrix} \text{ for } t \neq 1, \ t \neq -1$$

Determinants and their properties

A system $A\underline{x} = \underline{b}$ has either

① Unique solution

② no solution

③ Infinitely many solutions

When A is square, the determinant distinguishes case ① from the other cases

$$\det(A) \begin{cases} = 0 & \to \text{don't have a unique solution} \\ \neq 0 & \to \text{have a unique solution} \end{cases}$$

Consider 2×2 case :

① $ax + by = u$

② $cx + dy = v$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} u \\ v \end{pmatrix}$$

$d \times$ ① $\quad adx + bdy = du$

$b \times$ ② $\quad bcx + bdy = bv$

Subtract

$(ad - bc)x = du - bv$

If $ad - bc \neq 0$ then $x = \dfrac{du - bv}{ad - bc}$

$\uparrow$ unique solution iff $ad - bc \neq 0 \Rightarrow ad - bc = \det\begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$c \times$ ① $\quad acx + bcy = cu$

$a \times$ ② $\quad acx + ady = av$

Subtract

$(bc - ad)y = cu - av$

If $bc - ad \neq 0$, then $y = \dfrac{cu - av}{bc - ad}$

$\Rightarrow$ The determinant of a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$.

Consider 1×1 case

$(a)(x) = (u)$

$\to$ If $a \neq 0$, $x = u/a$

$\therefore \det(A) = a$ if $A = (a) \in Mat_{1,1}(\mathbb{R})$

Consider $n \times n$ case $\quad \Gamma$ = Matrix of Cofactors of A

$A = (a_{ij})$. Let M_{ij} be the $(n-1)\times(n-1)$ matrix obtained by crossing out the ith row and the jth column.

e.g. If $A = \begin{pmatrix} 2 & 3 & 1 \\ -2 & 3 & 5 \\ 6 & 7 & -3 \end{pmatrix}$

then $M_{13} = \begin{pmatrix} -2 & 3 \\ 6 & 7 \end{pmatrix}$

$M_{21} = \begin{pmatrix} 3 & 1 \\ 7 & -3 \end{pmatrix}$

In general, $\det A = |A| = \displaystyle\sum_{j=1}^{n} (-1)^{i+j} a_{ij} \cdot \det(M_{ij})$ $\quad$ (i fixed - choose $i = 1, 2, \dots, n$ - formula is independent of i.)

$$= (-1)^{i+1} a_{i1} \det(M_{i1}) + (-1)^{i+2} a_{i2} \det(M_{i2}) + \dots + (-1)^{i+n} a_{in} \det(M_{in})$$

Example: $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $i=2$:

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = (-1)^{2+1} cb + (-1)^{2+2} da = -bc + ad = ad - bc$$

3x3 example:

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}, \quad \text{Choosing } i=1$$

$$\det \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$= a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$$

4x4 example:

$$\begin{pmatrix} 2 & 0 & 3 & 1 \\ 0 & 2 & 3 & 5 \\ 0 & 1 & 2 & 4 \\ 1 & 0 & 0 & 2 \end{pmatrix} \quad \text{Choose } i=4$$

$$\det \begin{pmatrix} 2 & 0 & 3 & 1 \\ 0 & 2 & 3 & 5 \\ 0 & 1 & 2 & 4 \\ 1 & 0 & 0 & 2 \end{pmatrix} = -1 \begin{vmatrix} 0 & 3 & 1 \\ 2 & 3 & 5 \\ 1 & 2 & 4 \end{vmatrix} + 0 - 0 + 2 \begin{vmatrix} 2 & 0 & 3 \\ 0 & 2 & 3 \\ 0 & 1 & 2 \end{vmatrix}$$

(etc)

3x3 Example 2: The Sequel

$$A = \begin{pmatrix} 2 & 3 & 4 \\ 2 & 0 & 1 \\ 0 & 3 & 2 \end{pmatrix}$$

With $i=1$

$$|A| = 2\begin{vmatrix} 0 & 1 \\ 3 & 2 \end{vmatrix} - 3\begin{vmatrix} 2 & 1 \\ 0 & 2 \end{vmatrix} + 4\begin{vmatrix} 2 & 0 \\ 0 & 3 \end{vmatrix}$$

$$= 2\times -3 \; -3\times 4 + 4\times 6 = 6$$

doesn't matter what i you use ☺

With $i=3$

$$|A| = 0 - 3\begin{vmatrix} 2 & 4 \\ 2 & 1 \end{vmatrix} + 2\begin{vmatrix} 2 & 3 \\ 2 & 0 \end{vmatrix}$$

$$= 0 - 3\times(-6) + 2\times(-6) = 6$$

Determinant Properties:

(i) The value of the determinants do not depend on which row we use for the cofactor expansion

(ii) If $A \in Mat_{n\times n}(\mathbb{R})$, then $\det(A^T) = \det(A)$

(iii) $\det(I_n) = 1$

(iv) $\det\begin{pmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & a_{nn} \end{pmatrix} = a_{11}a_{22}\cdots a_{nn}$, $\det\begin{pmatrix} a_{11} & * & \cdots & * \\ & a_{22} & \ddots & \vdots \\ & & \ddots & * \\ 0 & & & a_{nn} \end{pmatrix} = a_{11}a_{22}\cdots a_{nn}$

diagonal matrix

upper triangular matrix

$\det\begin{pmatrix} a_{11} & & & 0 \\ * & a_{22} & & \\ \vdots & \ddots & \ddots & \\ * & \cdots & * & a_{nn} \end{pmatrix} = a_{11}a_{22}\cdots a_{nn}$

lower triangular matrix

just multiply the diagonals together ☺

(v) For $A, B \in Mat_{n\times n}(\mathbb{R})$,

$\det(AB) = \det(A) \times \det(B)$

(vi) We can apply EROs:

- doing a swap $R_i \leftrightarrow R_j$ of 2 rows of an $n\times n$ matrix A to produce a matrix A', then $\det(A') = -\det(A)$
- Multiplying a row R_i by a constant $c \neq 0$ causes determinant to be multiplied by c.
- Adding a scalar multiple of one row to another does not change the determinant.

Warning: $\det(A+B) \neq \det A + \det B$

Example

$$\begin{vmatrix} 1 & 2 & 3 \\ 2 & 0 & 1 \\ 0 & 1 & 0 \end{vmatrix} \overset{R_2-2R_1}{=} \begin{vmatrix} 1 & 2 & 3 \\ 0 & -4 & -5 \\ 0 & 1 & 0 \end{vmatrix} \overset{R_2 \leftrightarrow R_3}{=} -\begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & -4 & -5 \end{vmatrix} \overset{R_3+4R_2}{=} -\begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & -5 \end{vmatrix} = -(1\times 1\times -5) = 5$$

Or just do $0 - 1\begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix} + 0$

$= -(1-6) = 5$

Example:

$$\begin{vmatrix} 1 & 2 & 1 & 1 \\ 3 & 1 & 2 & 5 \\ -2 & 0 & 3 & 2 \\ 0 & 1 & -2 & 1 \end{vmatrix} \overset{\text{Several EROs}}{=} -13 \begin{vmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 7 \end{vmatrix} = -13\times 1\times 1\times 1\times 7 = -91$$

Inverse Matrices

For real numbers, if $x \neq 0$, then $x \times x^{-1} = 1$

Recall $I = I_n = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix}$ behaves like 1 in the matrix world, i.e. if $A \in Mat_{n\times n}(\mathbb{R})$, then $AI_n = I_nA = A$

(Square matrices only)

Definition: Given an $n\times n$ matrix A, the inverse A^{-1} of A is another $n\times n$ matrix satisfying:

$A \cdot A^{-1} = I_n = A^{-1}A$

But $A\cdot A^{-1} = I_n$ iff $A^{-1}A = I_n$ 🙂 $\rightarrow$ do not have to check both ways

If A^{-1} exists, we say A is invertible. Otherwise A is not invertible.

(non-singular) (Singular)

When looking at 2×2 determinants, we solved:

$ax + by = u$

$cx + dy = v$

given $ad - bc \neq 0$

(i.e. $\det A \neq 0$)

let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} u \\ v \end{pmatrix} \rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \dfrac{du-bv}{ad-bc} \\ \dfrac{av-cu}{ad-bc} \end{pmatrix} = \begin{pmatrix} \dfrac{d}{ad-bc} & \dfrac{-b}{ad-bc} \\ \dfrac{a}{ad-bc} & \dfrac{-c}{ad-bc} \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix} = \frac{1}{\det A}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix}$$

Unique Solution

$A\underline{x} = \underline{u}$ $\qquad \underline{x} = \begin{pmatrix} x \\ y \end{pmatrix}$ $\quad \underline{u} = \begin{pmatrix} u \\ v \end{pmatrix}$

$$\underline{x} = \frac{1}{\det A}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \cdot \underline{u}$$
$$= A^{-1}$$

$$\underline{x} = \frac{1}{\det A}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix}\underline{x}$$

$$\underline{x} = \frac{1}{\det A}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \cdot A\underline{x}$$

$$\frac{1}{\det A}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \cdot A = \underline{I}_2$$

$$\therefore \frac{1}{\det A}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = A^{-1}$$

(can also multiply A by A^{-1} to get I_2)

Example of 2×2 matrix with no inverse

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}\begin{pmatrix} p & q \\ r & s \end{pmatrix} = \begin{pmatrix} p+2r & q+2s \\ 2p+4r & 2q+4s \end{pmatrix}$$

requires $p+2r=1 \rightarrow 2p+4r=2$
$2p+4r=0$ $\Big\}$ Contradiction

$$\det\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} = 1\times 4 - 2\times 2 = 0$$

Theorem: A square matrix $A \in Mat_n(\mathbb{R})$ is invertible iff $\det A \neq 0$.

If A^{-1} exists, then $A \cdot A^{-1} = I_n$

$\Rightarrow \det(A \cdot A^{-1}) = \det(I_n) = 1$

$\det A \cdot \det A^{-1} = 1$

$\hookrightarrow$ requiring $\det A \neq 0$

If $A \in Mat_n(\mathbb{R})$ is invertible, then $\det(A) \neq 0$

$1 = \det(I_n) = \det(A \cdot A^{-1}) = \det A \times \det A^{-1} \rightarrow$ both $\det A$, $\det(A^{-1}) \neq 0$

Compute inverses using Gaussian elimination:

Let $A = Mat_n(\mathbb{R})$ with $\det A \neq 0$

Write a big augmented matrix

$$\left(A \mid I_n\right)$$

Apply EROs on both sides until A becomes I_n

$$\left(A \mid I_n\right) \xrightarrow[\text{EROs}]{\text{lots of}} \left(I_n \mid B\right)$$

Then $B = A^{-1}$ (as long as $\det A \neq 0$)

Example: $A = \begin{pmatrix} 2 & 5 & -4 \\ 1 & 1 & 2 \\ 1 & 2 & -1 \end{pmatrix}$

$$\left(\begin{array}{ccc|ccc} 2 & 5 & -4 & 1 & 0 & 0 \\ 1 & 1 & 2 & 0 & 1 & 0 \\ 1 & 2 & -1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_3 \leftrightarrow R_1} \left(\begin{array}{ccc|ccc} 1 & 2 & -1 & 0 & 0 & 1 \\ 1 & 1 & 2 & 0 & 1 & 0 \\ 2 & 5 & -4 & 1 & 0 & 0 \end{array}\right) \xrightarrow[R_3 - 2R_1]{R_2 - R_1} \left(\begin{array}{ccc|ccc} 1 & 2 & -1 & 0 & 0 & 1 \\ 0 & -1 & 3 & 0 & 1 & -1 \\ 0 & 1 & -2 & 1 & 0 & -2 \end{array}\right)$$

$$\xrightarrow{-R_2} \left(\begin{array}{ccc|ccc} 1 & 2 & -1 & 0 & 0 & 1 \\ 0 & 1 & -3 & 0 & -1 & 1 \\ 0 & 1 & -2 & 1 & 0 & -2 \end{array}\right) \xrightarrow[R_3 - R_2]{R_1 - 2R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 5 & 0 & 2 & -1 \\ 0 & 1 & -3 & 0 & -1 & 1 \\ 0 & 0 & 1 & 1 & 1 & -3 \end{array}\right) \xrightarrow[R_2 + 3R_3]{R_1 - 5R_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -5 & -3 & 14 \\ 0 & 1 & 0 & 3 & 2 & -8 \\ 0 & 0 & 1 & 1 & 1 & -3 \end{array}\right)$$

$$\therefore A^{-1} = \begin{pmatrix} -5 & -3 & 14 \\ 3 & 2 & -8 \\ 1 & 1 & -3 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 5 & -4 \\ 1 & 1 & 2 \\ 1 & 2 & -1 \end{pmatrix}\begin{pmatrix} -5 & -3 & 14 \\ 3 & 2 & -8 \\ 1 & 1 & -3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

If the process to compute A^{-1} using EROs breaks, i.e. cannot get I_n on LHS, then A is not invertible so $\det A = 0$.

Computing inverses using the Cofactor method:

Given $n \times n$ matrix A with $\det A \neq 0$

Compute the $C_{ij} = (-1)^{i+j} \det(M_{ij})$ where M_{ij} = matrix A after removing the i^{th} row and the j^{th} column.

Let C be the matrix of C_{ij}, i.e. $C = \begin{pmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{pmatrix}$

Then $A^{-1} = \dfrac{1}{\det A} C^T$

(if $\det A = 0$, we end up dividing by 0 = bad)

$A = \begin{pmatrix} 2 & 5 & -4 \\ 1 & 1 & 2 \\ 1 & 2 & -1 \end{pmatrix}$

$C_{11} = \begin{vmatrix} 1 & 2 \\ 2 & -1 \end{vmatrix} = -5$ $\qquad C_{12} = -\begin{vmatrix} 1 & 2 \\ 1 & -1 \end{vmatrix} = 3$ $\qquad C_{13} = \begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix} = 1$

$C_{21} = -\begin{vmatrix} 5 & -4 \\ 2 & -1 \end{vmatrix} = -3$ $\qquad C_{22} = \begin{vmatrix} 2 & -4 \\ 1 & 1 \end{vmatrix} = 2$ $\qquad C_{23} = -\begin{vmatrix} 2 & 5 \\ 1 & 2 \end{vmatrix} = 1$

$C_{31} = 14$ $\qquad C_{32} = -8$ $\qquad C_{33} = -3$

$\det A = 1$

$$A^{-1} = \frac{1}{\det A} C^T = \frac{1}{1}\begin{pmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{pmatrix} = \begin{pmatrix} -5 & -3 & 14 \\ 3 & 2 & -8 \\ 1 & 1 & -3 \end{pmatrix}$$

If we have $A\underline{x} = \underline{b}$, $A \in Mat_n(\mathbb{R})$:

If $\det A \neq 0$ then $A^{-1} \in Mat_n(\mathbb{R})$ exists

i.e. $A^{-1}(A\underline{x}) = A^{-1}\underline{b}$

$(A^{-1}A)\underline{x} = A^{-1}\underline{b}$

$I_n \underline{x} = \underline{x}$

$\therefore \underline{x} = A^{-1}\underline{b}$

Example: Solve $\begin{cases} 2x+5y-4z=-13 \\ x+y+2z=7 \\ x+2y-z=-3 \end{cases}$

$$\underline{b} = \begin{pmatrix} -13 \\ 7 \\ -3 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^{-1}\underline{b} = \begin{pmatrix} -5 & -3 & 14 \\ 3 & 2 & -8 \\ 1 & 1 & -3 \end{pmatrix}\begin{pmatrix} -13 \\ 7 \\ -3 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$$

More on Determinants and Inverses

$A \in Mat_n(\mathbb{R})$ is invertible iff $\det A \neq 0$

$(A|I_n) \xrightarrow{EROs} (I_n|B)$ if this process is successful then $B = A^{-1}$

S: Swap $R_i \leftrightarrow R_j$

M: Multiply λR_i $(\lambda \neq 0)$

A: Add $R_i + \mu R_j$ $(\mu \in \mathbb{R})$

EROs as matrix multiplications

$A \overset{ERO}{\rightarrow} E * A$ Same size elementary matrix E

- $R_i \leftrightarrow R_j$ Swap row i with row j. Let P_{ij} be the identity matrix with the ith and jth rows swapped. Then this row operation is $A \rightarrow P_{ij}A$

e.g. $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 2 & 3 \\ 7 & 8 & 9 \\ 4 & 5 & 6 \end{pmatrix}$ =

$$P_{23} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$P_{23}A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 7 & 8 & 9 \\ 4 & 5 & 6 \end{pmatrix}$$

– λR_i, $\lambda \neq 0$ Multiply ith row by λ. Let $M_i(\lambda)$ be the identity matrix with the i^{th} row multiplied by λ.
Then this row operation is $A \xrightarrow{\lambda R_i} A' = M_i(\lambda) \times A$

E.g.

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \xrightarrow{3R_2} \begin{pmatrix} 1 & 2 & 3 \\ 12 & 15 & 18 \\ 7 & 8 & 9 \end{pmatrix}$$

$=$

$$M_2(3) \times A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 12 & 15 & 18 \\ 7 & 8 & 9 \end{pmatrix}$$

– $R_i + \mu R_j$ Add μ times row j to row i. Let $A_{ij}(\mu)$ be the identity matrix but replace the $(i,j)^{th}$ entry with μ. Then this operation is $A \xrightarrow{R_i + \mu R_j} A' = A_{ij}(\mu) \times A$

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \xrightarrow{R_2 - 4R_1} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 7 & 8 & 9 \end{pmatrix}$$

$=$

$$A_{21}(-4) \times A = \begin{pmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 7 & 8 & 9 \end{pmatrix}$$

Suppose $\det A \neq 0$, so its RREF is I_n, i.e. A can be converted into I_n by a sequence of EROs:

$$I_n = E_N\left(\cdots E_3\left(E_2\left(E_1 \times A\right)\right)\cdots\right)$$

(E_i elementary matrices: P_{ij}, $M_i(\lambda)$, $A_{ij}(\mu)$)

$$(A \mid I_n) \xrightarrow{ERO_1} (E_1 \times A \mid E_1 \times I_n) \xrightarrow{ERO_2} \left(E_2(E_1 \times A) \mid E_2(E_1 \times A)\right) \rightarrow \cdots$$

$$\xrightarrow{ERO_N} \left(\underbrace{E_N(E_{N-1}(\cdots(E_1 \times A)\cdots))} \mid E_N(E_{N-1}(\cdots(E_1 \times I_n)\cdots))\right)$$

$E_N(\cdots(E_1 \times A)\cdots) = I_n$

$$\underbrace{(E_N \times E_{N-1} \times \cdots \times E_1)}_{= A^{-1}} \overset{\text{associativity}}{\times} A = I_n$$

Properties of inverses

Let $A \in Mat_n(\mathbb{R})$ be invertible (equiv. to $\det A \neq 0$), then there exists a unique matrix $B \in Mat_n(\mathbb{R})$ s.t. $A \times B = B \times A = I_n$

If B satisfies $A \times B = I_n$, then it automatically also satisfies $B \times A = I_n$ and vice versa

- Let $A, B \in Mat_n(\mathbb{R})$, and $\det A \neq 0$, $\det B \neq 0$

  $\Rightarrow \det(AB) = \underset{\neq 0}{\det A} \times \underset{\neq 0}{\det B} \neq 0$

  $\Rightarrow AB$ is invertible

- $(AB)^{-1} = B^{-1}A^{-1}$

  $I_n = (AB)(AB)^{-1} = (AB)(B^{-1}A^{-1}) \overset{\text{associativity}}{=} A \cdot (B \cdot B^{-1}) \cdot A^{-1}$
  $= A \cdot A^{-1}$
  $= I_n$

- If $A_1, A_2, \ldots, A_K \in Mat_n(\mathbb{R})$ are invertible, then $B = A_1 \times A_2 \times \ldots \times A_K \in Mat_n(\mathbb{R})$ is also invertible and we have $(A_1 \cdots A_K)^{-1} = A_K^{-1} \times A_{K-1}^{-1} \times \ldots \times A_1^{-1}$

- This is similar to transpose:

  $(A_1 \cdots A_K)^T = A_K^T \ldots A_1^T$

  Since $\det A = \det(A^T)$, invertibility of $A \Rightarrow$ invertibility of A^T

  Assume $A \in Mat_n(\mathbb{R})$ is invertible, then $(A^T)^{-1} = (A^{-1})^T$

- $(A^{-1})^{-1} = A$ if A invertible

Made with Goodnotes